WEEK 3 · DAY 2 · OVERVIEW

More Visualization



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ABOUT 60 MIN WITH EXERCISES / 8 SECTIONS / 7 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The Desk Lead's Note

Your seven charts went round the desk. Three asks before the meeting. Put a line through a scatter and give me its slope, in units. Show me how many bonds stand behind each shape. Then make the charts meeting-ready: our colours, numbers I can read, the bond everyone asks about marked, and a file for the slide. No chart says more than the data.

ABOUT 60 MIN WITH EXERCISES / 8 SECTIONS / 7 EXERCISES

Your Route Through Day 2

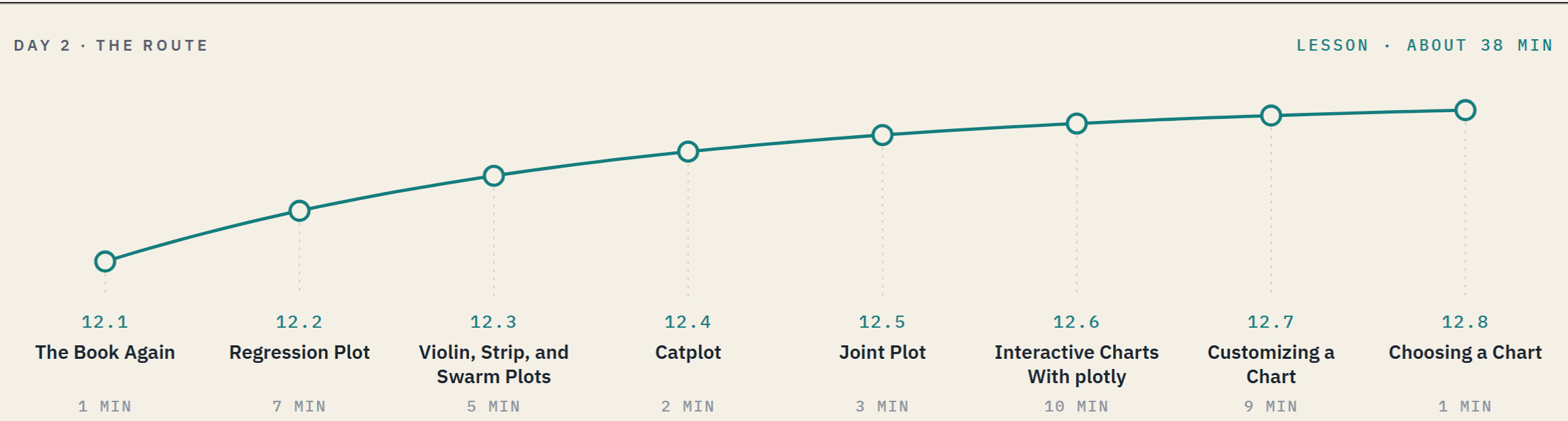


FIGURE 1 Eight sections, in order, with about how long each takes.

Time. About 38 minutes for the lesson, then about 21 for the seven exercises.

12.1 The Book Again

YESTERDAY'S BOOK IN ONE CELL

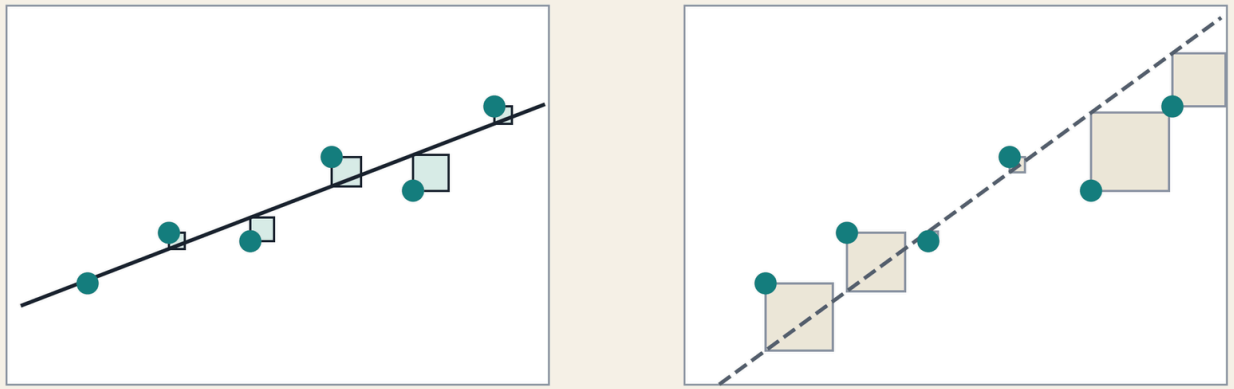
```
sns.set_theme(style='whitegrid', palette=['#2E7D5B', '#9AA5B1', '#1E4D3A', '#52606D'],
              rc={'boxplot.medianprops.color': '#1E4D3A'})
holdings = pd.read_csv(data_folder + 'holdings.csv')
ratings = pd.read_csv(data_folder + 'ratings.csv')
book = pd.merge(holdings, ratings, on='rating')
bucket_order = ratings.sort_values('score')['bucket'].unique().tolist()
print(book.shape)
```

```
(200, 22)
```

Same table, same credit order, same greens: today adds a slope, the bonds, and the finish.

12.2 Regression Plot

Least squares: the line whose squares add to the smallest total area



The fitted line: slope 0.50, total area 1.84

A steeper line: slope 0.94, total area 9.66

Each bond's vertical gap to the line, squared, is a square. np.polyfit(x, y, deg=1) finds the slope and intercept that make the green total smallest.

FIGURE 2 Each bond's gap to a line drawn as a square; least squares picks the line with the smallest total area.

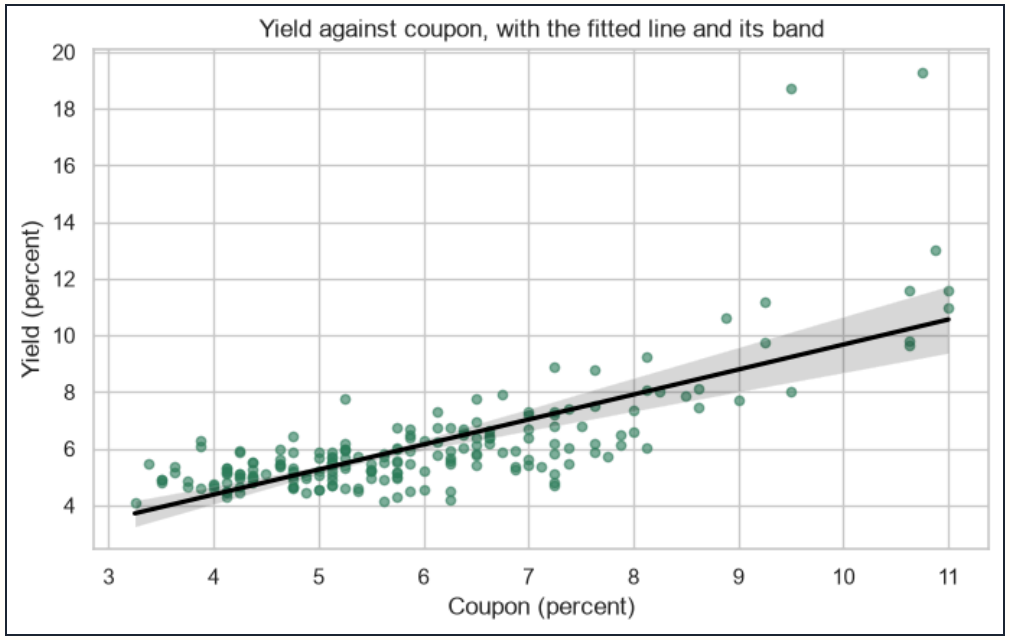
EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
Slope	`np.polyfit(x, y, deg=1)[0]`, 0.88	`=SLOPE(L2:L201, G2:G201)`
Intercept	`np.polyfit(x, y, deg=1)[1]`, 0.85	`=INTERCEPT(L2:L201, G2:G201)`
R squared	`x.corr(y) ** 2`, 0.55	`=RSQ(L2:L201, G2:G201)`
The line on the chart	`sns.regplot()`	Chart Elements, Trendline, Linear

Slope, intercept, R squared, each with its unit: print them beside any line.

The Fit and Its Band

WHAT REGPLOT DRAWS



FIT THE LINE

```
slope, intercept = np.polyfit(book['coupon'], book['yield_pct'], deg=1)
print('Slope:', round(slope, 2), 'points of yield per point of coupon')
print('Intercept:', round(intercept, 2), 'percent')
print('The line at 5 and 8 percent coupons:',
      round(intercept + slope * 5, 2), round(intercept + slope * 8, 2))
print('Lowest coupon in the book:', book['coupon'].min())
```

```
Slope: 0.88 points of yield per point of coupon
Intercept: 0.85 percent
The line at 5 and 8 percent coupons: 5.26 7.91
Lowest coupon in the book: 3.25
```

12.3 Violin, Strip, and Swarm Plots

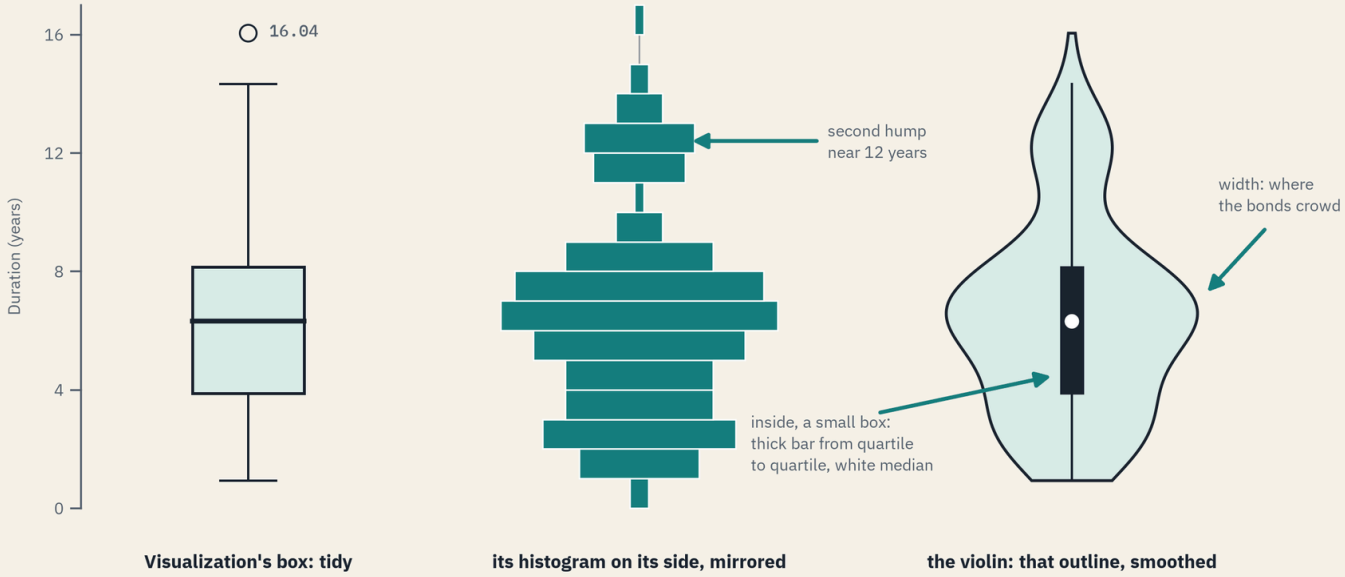


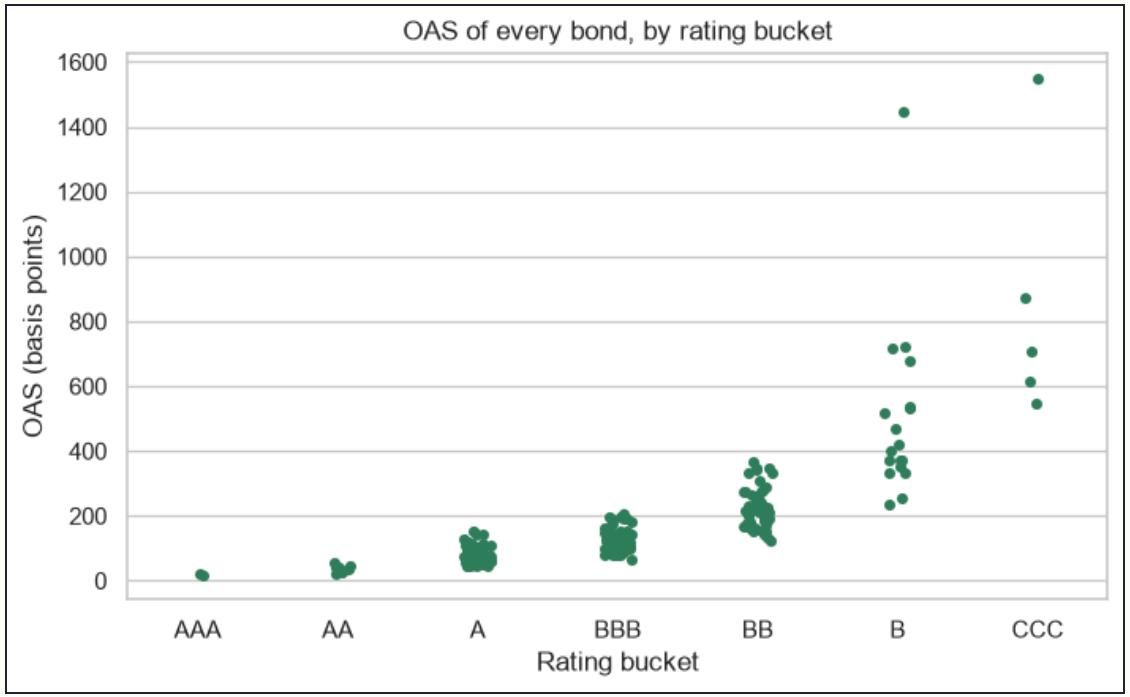
FIGURE 3 Duration three ways: Visualization's box, the histogram on its side and mirrored, and the violin that smooths it.

EXCEL SIDE BY SIDE

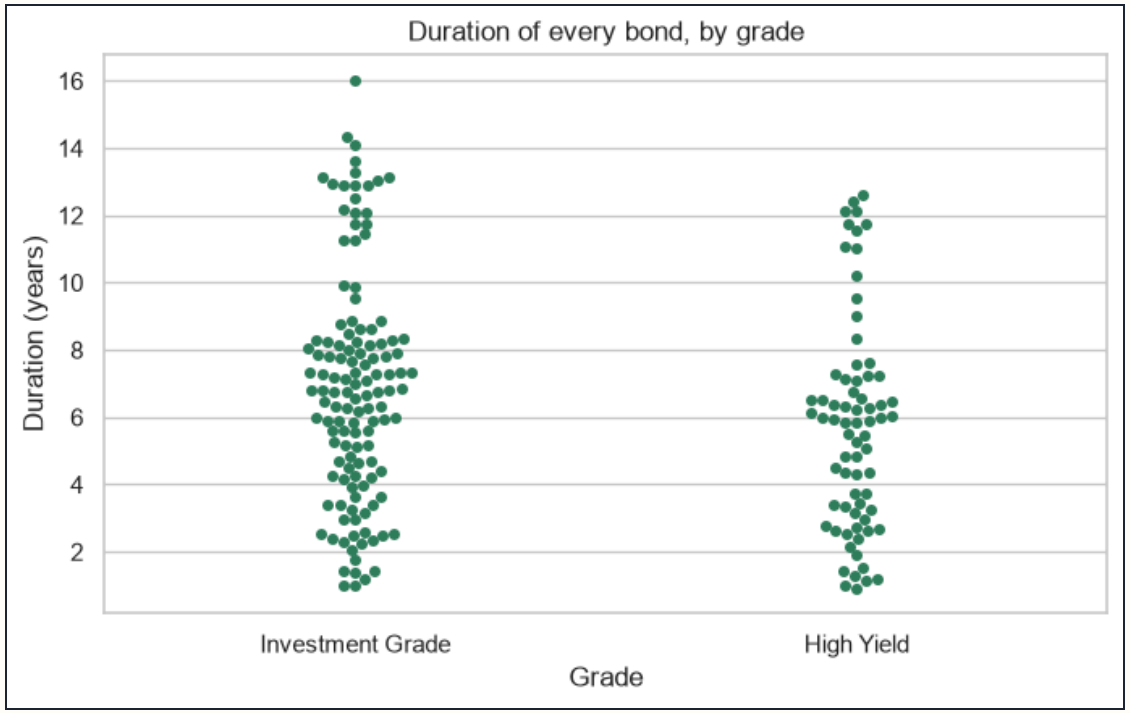
JOB	PYTHON	EXCEL
Shape of each group	`sns.violinplot()`	no violin chart; Box and Whisker is nearest
Every bond as a dot	`sns.stripplot()`	Box and Whisker, Format Data Series, Show inner points
One group's lowest	19	`=MINIFS(M:M, V:V, "Investment Grade")`; `MAXIFS` gives 204

Every Bond as a Dot

A DOT PER BOND, NUDGED



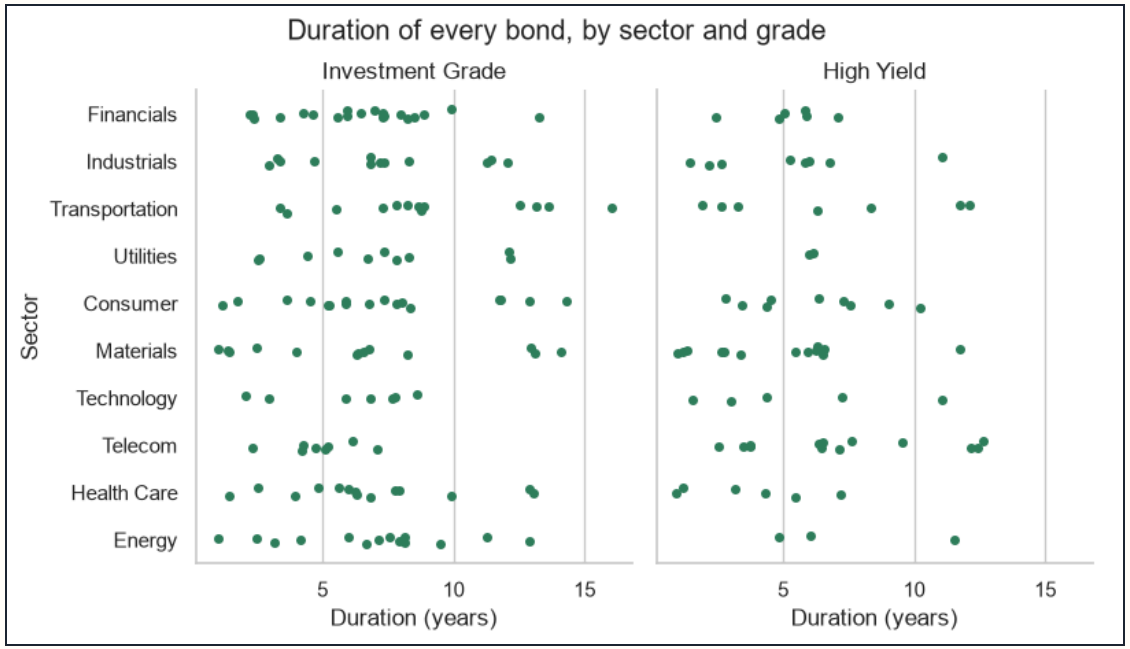
A DOT PER BOND, PACKED



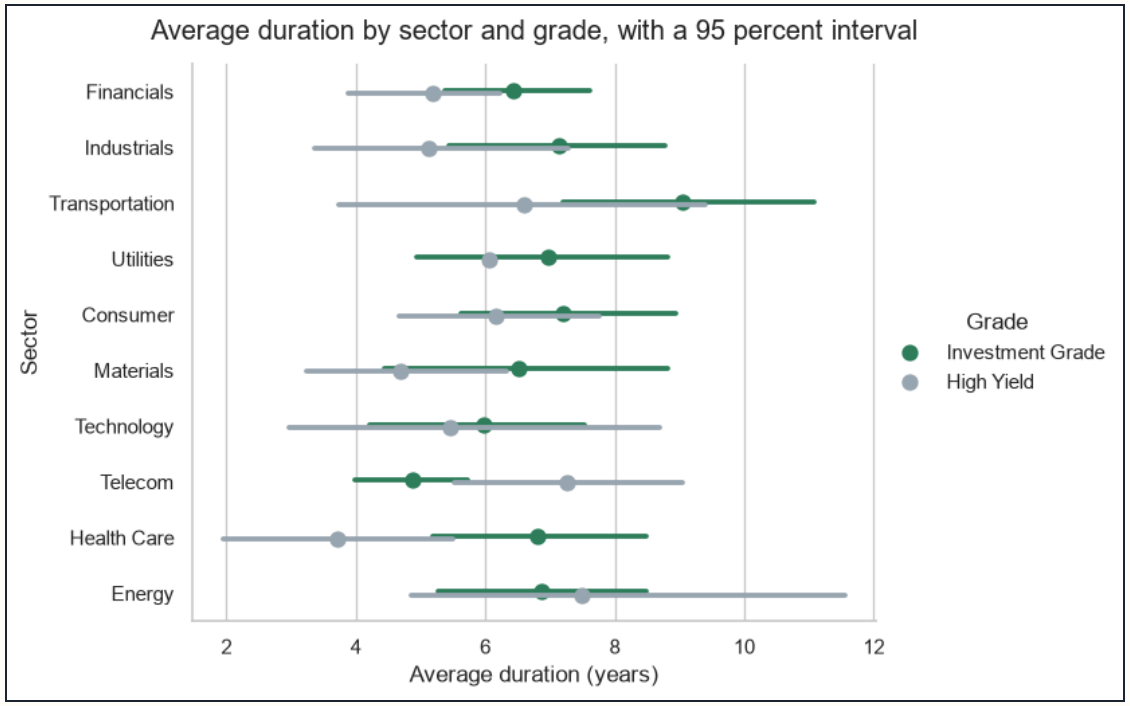
Dots show how many bonds stand behind a shape, and one function draws all of these.

12.4 Catplot

A PANEL PER GRADE



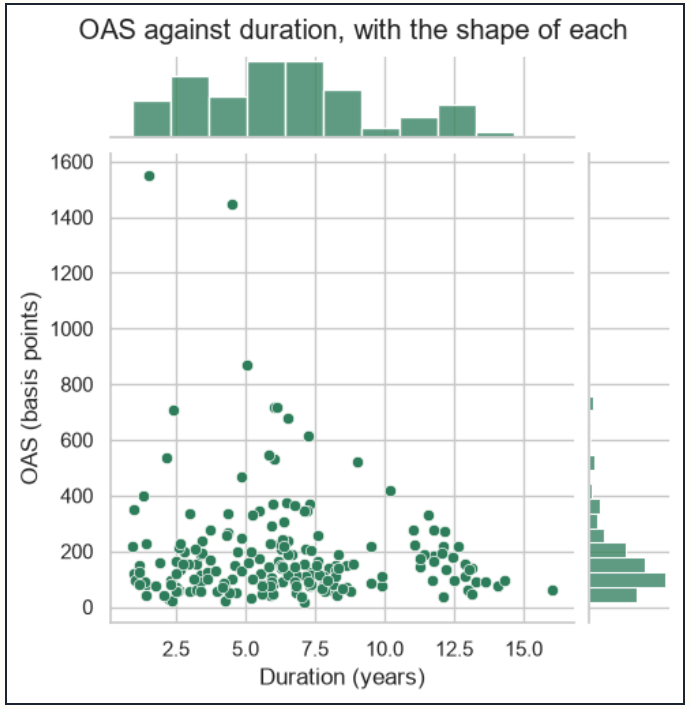
AVERAGES AND THEIR RANGE



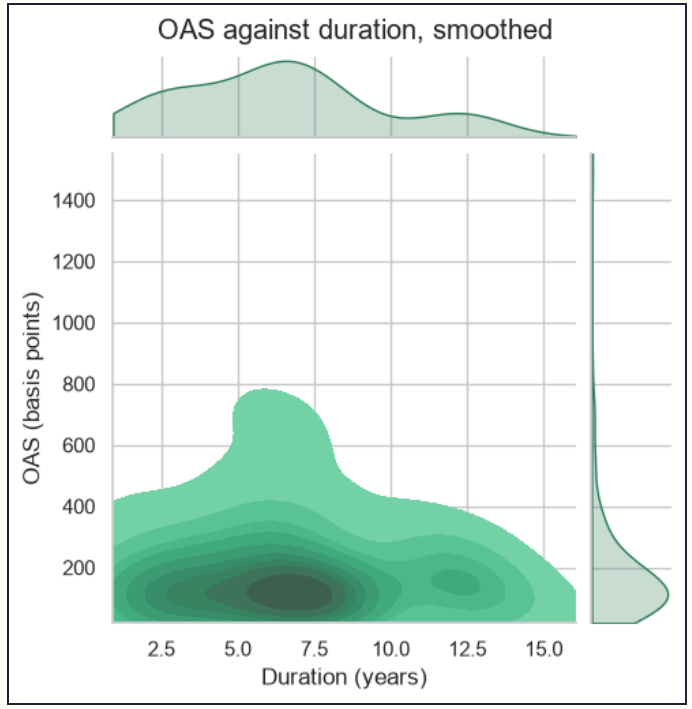
Averages say where the middle is; the dots say how many stand behind it. Back to one scatter, and how crowded it is.

12.5 Joint Plot

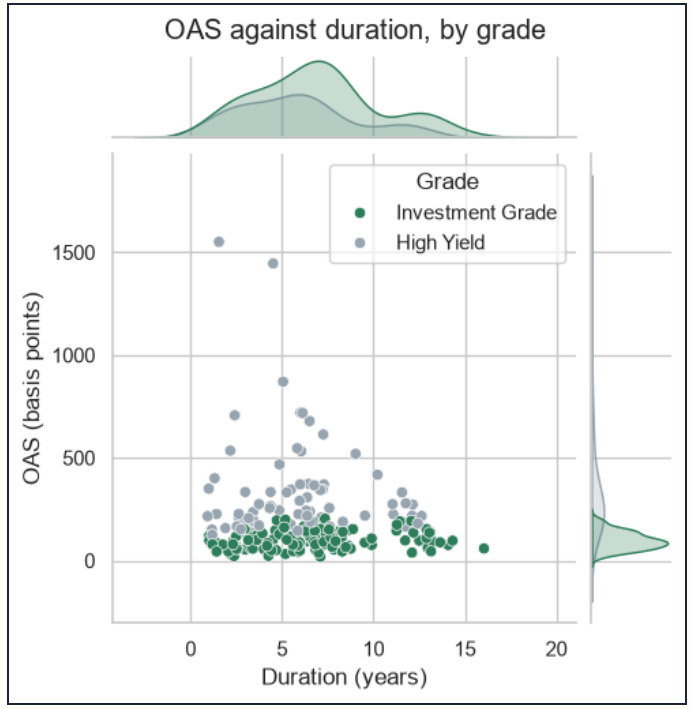
SCATTER AND SHAPES



SMOOTHED



ONE CURVE PER GRADE



A crowd hides which bond is which. Next, a chart that names it when you point.

12.6 Interactive Charts With plotly

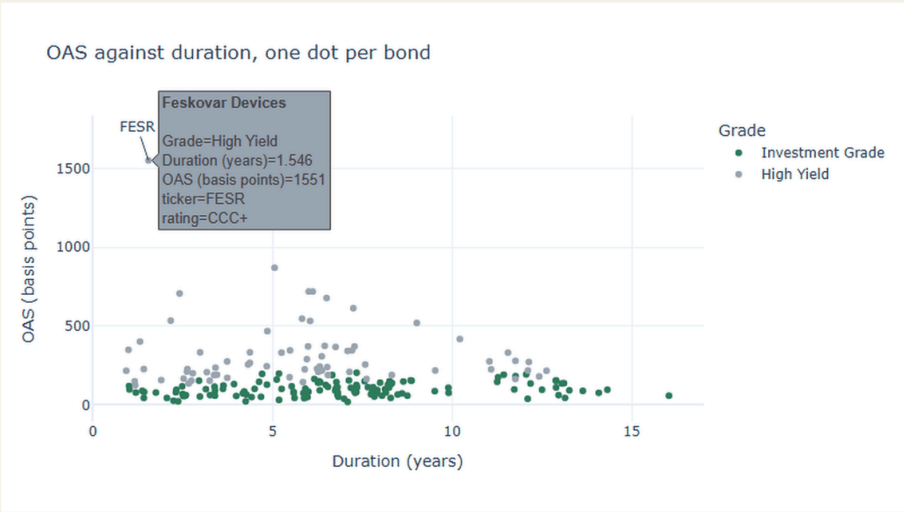


FIGURE 4 The plotly scatter of OAS against duration, the hover box on the highest dot: Feskovar Devices, FESR.

THE TOP DOT

The top dot is Feskovar Devices, an invented issuer: FESR, rated CCC+, at 1.546 years and 1,551 bp.

Point at a box for its quartiles, or at a dot for its bond: Visualization's circles now have names.

A Surface From a Formula

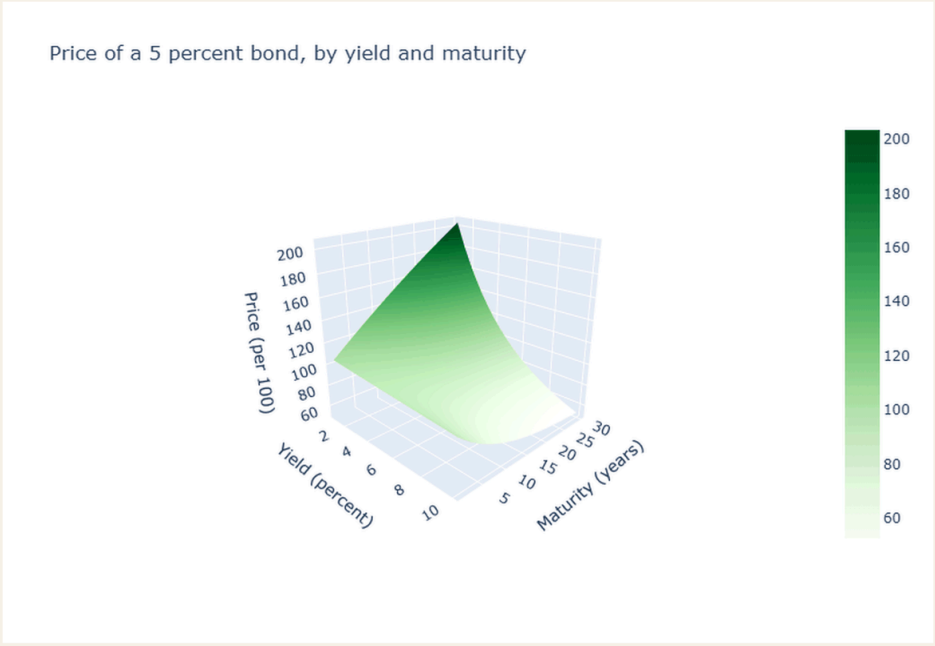


FIGURE 8 The plotly surface of a 5 percent bond's price over yield and maturity, level at 100 along the 5 percent line.

EVERY PAIR OF TWO LISTS

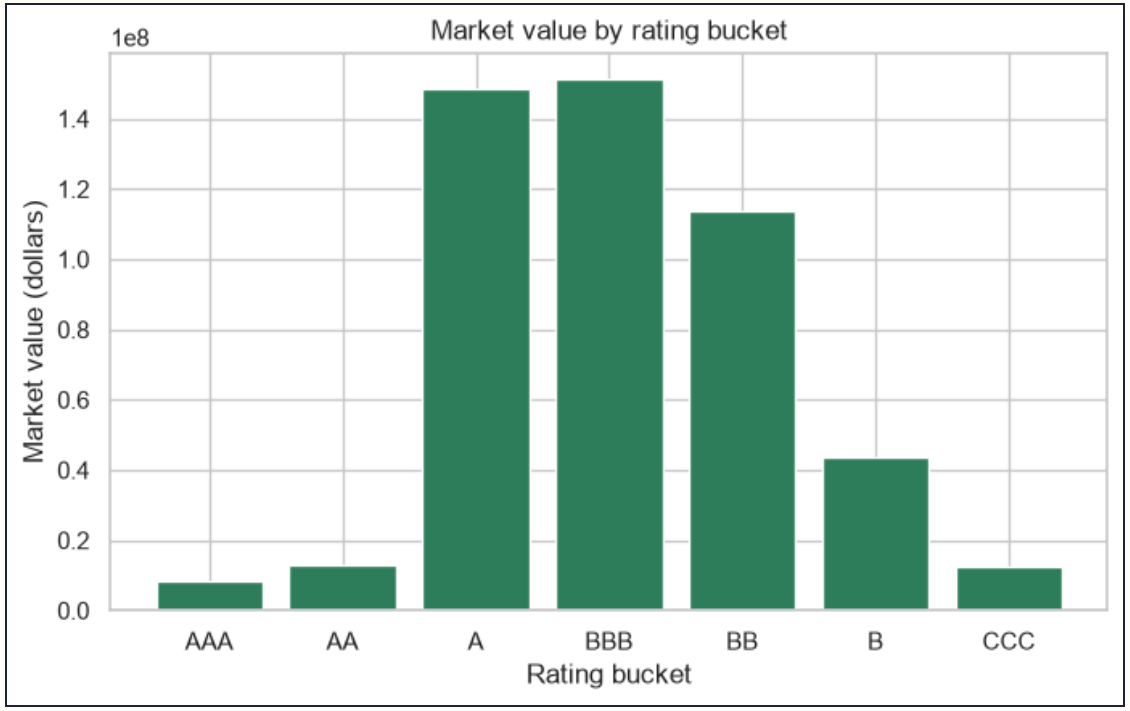
```
print(small_yields)
print(small_years)
```

```
[[4 6]
 [4 6]
 [4 6]]
[[1 1]
 [2 2]
 [3 3]]
```

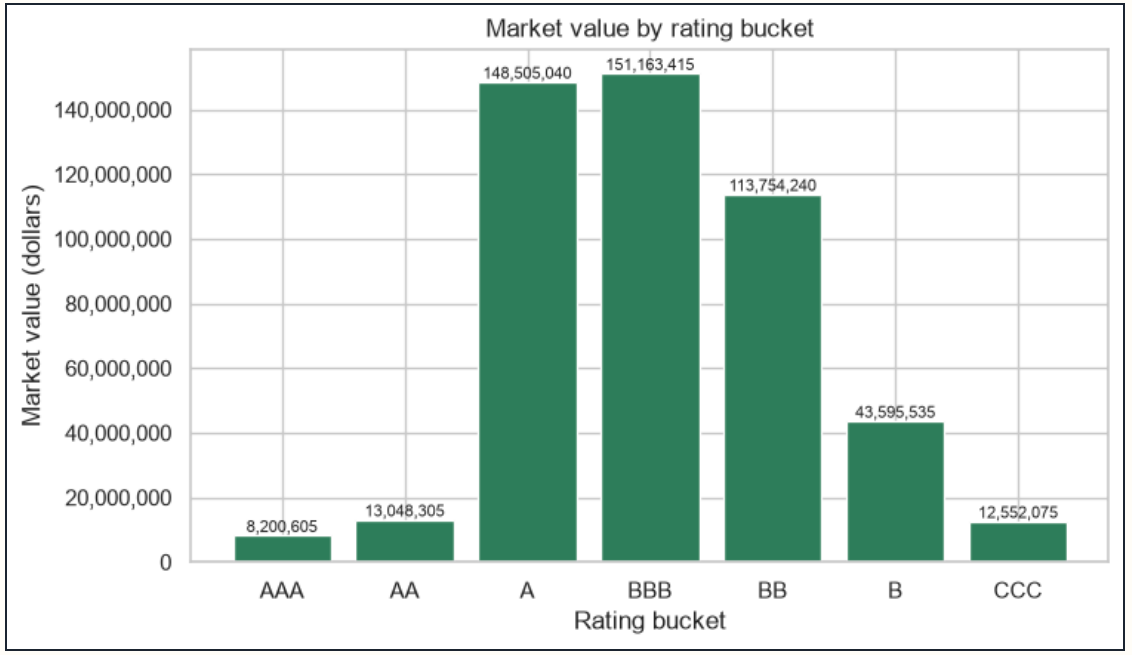
On screen a chart can answer the mouse; a meeting still wants a still picture in the desk's look.

Dollars a Reader Can Read

BEFORE

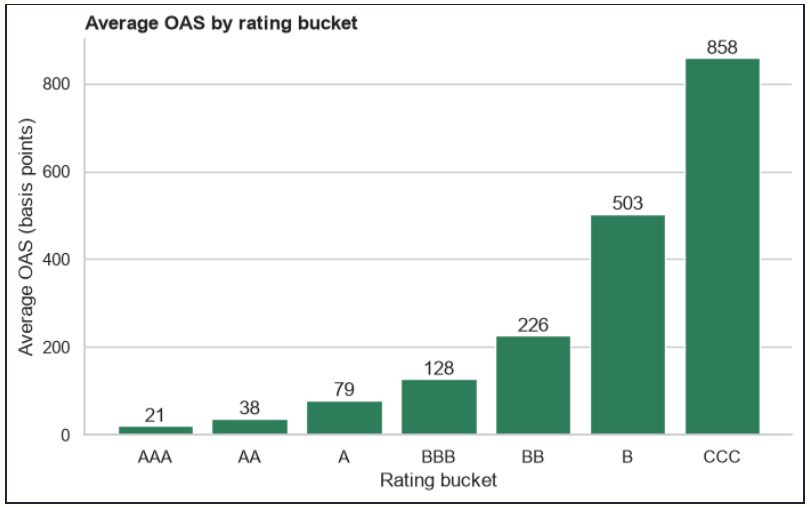


AFTER



12.7 Customizing a Chart

ONE DESK STYLE, SAVED



EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
Axis limits	`ax.set_ylim(0, 600)`	Format Axis, Bounds, Minimum and Maximum
Thousands separators	`ax.yaxis.set_major_formatter('{x:,.0f}')`	Format Axis, Number, Format Code `#,##0`
A value on each bar	`ax.bar_label(bars)`	Chart Elements, Data Labels
Save as a picture	`fig.savefig('name.png')`	right-click the chart, Save as Picture

One function holds the look; one line writes the file.

12.8 Choosing a Chart

Today's charts, and the question each one answers

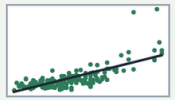
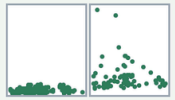
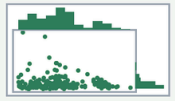
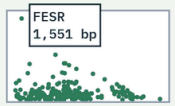
The question	Today's chart	Its name	Python	Excel
How much y moves with x		**Regression plot**	`np.polyfit` `sns.regplot, sns.lmplot`	Trendline, SLOPE
Each group's shape		**Violin plot**	`sns.violinplot`	Box and Whisker, nearest
Every bond in each group		**Strip, swarm**	`sns.stripplot` `sns.swarmplot`	Show inner points
One chart per group		**Panels**	`sns.catplot(col=)` `plt.subplots(1, 2)`	a chart per group
Two columns, and each one's shape		**Joint plot**	`sns.jointplot` `ax.hexbin`	a Scatter, two Histograms
Which bond is that dot?		**Interactive**	`px.scatter, px.box` `px.bar`	the hover tip
A height from two inputs		**3D, surface**	`px.scatter_3d` `go.Surface`	3-D Surface

FIGURE 9 Seven questions, each with today's chart, its calls, and its nearest Excel feature.

You Can Now

- Fit a line, give its slope, intercept, and R squared in units, and draw it with **`regplot()`** or **`lmplot()`**.
- Show each group's shape with a violin and every bond with a strip or swarm.
- Split a chart into panels with **`catplot()`**, and draw a joint plot.
- Draw a plotly chart, read a bond off it, and save it as a web page.
- Draw a 3D scatter and a surface, and say when 3D misleads.
- Finish a chart for a meeting, in one style, saved to a file.

NEXT
WEEK 3 · DAY 3

Exploratory Data Analysis

A holdings file arrives with problems in it. Find them by method, decide what to do about each, and show what every decision does to the numbers.